

?What does it mean that C# is Managed Code -5

(Managed Code means that C# code runs under the control of the .NET Runtime (CLR

:The CLR manages important things such as

- Memory management
- Garbage Collection
- Exception handling
- Type safety
- Code execution

.So, the programmer doesn't need to manually manage memory in most cases

?What does it mean that a **struct** is like a **class** -6

:A **struct** is similar to a **class** because both can contain

- Fields
- Properties
- Methods
- Constructors

:However, the main difference is

- **Class** → **Reference Type**
- **Struct** → **Value Type**

So, a **struct** can have many of the same members as a class, but it behaves differently when it is assigned or passed around because its **value is copied**

firstName - _firstName : self //

firstName → غالباً اسم متغير عادي، ويكون **local variable** أو parameter حسب المكان.
_firstName → غالباً **private field** داخل الـ class، والـ _ مجرد naming convention عشان تميّزه كـ field.
FirstName → غالباً **Property**، ودي اللي بتتعامل معاها من خارج الـ class

#How to Avoid Memory Leaks in C

C# uses **Garbage Collection (GC)** to automatically manage memory, but memory leaks can still happen when objects are kept referenced unnecessarily

:To avoid memory leaks

- **Unsubscribe from events** when they are no longer needed

- **.Dispose unmanaged resources** using **IDisposable** and **using**
- **.Avoid unnecessary static references**
- **.Stop and dispose timers** when they are no longer needed
- **.Remove unnecessary references** to objects that should be collected

In short: Make sure objects and resources are no longer referenced when you finish using them, so the **Garbage Collector** can clean them up